BioSpecInfo

Software Bill of Materials (SBOM)

Samuele Pio Provenzano

Versione **bsi-v220** · commit `784d87b`

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Software Bill of Materials (SBOM)

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Application version	`bsi-v220`
Commit	`784d87bfb90ba5015cc3136b419f5447d5cb7145`
Generated (UTC)	`2026-10-16T12:22:25.158Z`
Machine-readable format	`../evidence/sbom.cdx.json` — CycloneDX 1.5

> The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	`RDKit_minimal.js`	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	`RDKit_minimal.wasm`	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	`three.min.js`	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	`three_bloom.js`	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — SkeletonUtils _{mrdoob e collaboratori}	`skeleton_utils.js`	MIT	12 kB	Copies of the animated enemies with their skeleton (ATOMIC PROTOCOL).

Component	Version/File	Licence	Size	Role
three.js — GLTFLoader _{mrdoob e collaboratori}	`gltf_loader.js`	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	`3Dmol-min.js`	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).
SmilesDrawer _{Daniel Probst}	`smiles-drawer.min.js`	MIT	246 kB	2D structure drawing from SMILES.
PDF.js (build legacy) _{Mozilla Foundation}	`vendor/pdfjs/pdf.min.mjs`	Apache-2.0	511 kB	Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet.
PDF.js — worker _{Mozilla Foundation}	`vendor/pdfjs/pdf.worker.min.mjs`	Apache-2.0	1290 kB	The PDF decoding work, off the main thread: a complex page does not freeze the interface.
sql.js (JS glue) _{sql.js / SQLite}	`lib/sql-wasm.js`	MIT	45 kB	In-browser SQLite database for the data exercises.
SQLite (WebAssembly) _{SQLite Consortium}	`lib/sql-wasm.wasm`	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	`lib/dimuon.js`	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.
**Fists (2023)	First Person Animations (adattato)** _{1Matzh (Sketchfab)}	`models/braccia-atomic.glb`	CC-BY-4.0	6530 kB

Component	Version/File	Licence	Size	Role
Robotic Hand (modello 3D) _{SeanNicolas (Sketchfab)}	`models/mano-robotica-atomic.glb`	CC-BY-4.0	838 kB	The robotic left hand, over which the element power lights up.
Small Lab (adattato) _{Biohazard Rooms (Sketchfab)}	`models/mappe/small.glb`	CC-BY-4.0	1506 kB	ATOMIC PROTOCOL — laboratory (wings 1, 6).
Sci-fi lab (adattato) _{Agata Wilczek (Sketchfab)}	`models/mappe/scifi1.glb`	CC-BY-4.0	6683 kB	ATOMIC PROTOCOL — laboratory (wings 2, 7).
Flynn's Secret Lab [Tron Legacy] (adattato) _{jvouillon (Sketchfab)}	`models/mappe/flynn.glb`	CC-BY-4.0	2319 kB	ATOMIC PROTOCOL — laboratory (wings 3, 8).
Location underground laboratory (adattato) _{bazz4r (Sketchfab)}	`models/mappe/underground.glb`	CC-BY-4.0	10512 kB	ATOMIC PROTOCOL — laboratory (wings 4, 9).
Overgrown Laboratory (adattato) _{MichalTroye (Sketchfab)}	`models/mappe/overgrown.glb`	CC-BY-4.0	6227 kB	ATOMIC PROTOCOL — laboratory (wings 5, 10).
Quantum_Laboratory_X_Alpha (adattato) _{-X-ScornGames (Sketchfab)}	`models/mappe/reattore.glb`	CC-BY-4.0	3269 kB	ATOMIC PROTOCOL — apparatus (the reactor).
Sci Fi lab (adattato) _{IrisProcess (Sketchfab)}	`models/mappe/capsula.glb`	CC-BY-4.0	6548 kB	ATOMIC PROTOCOL — apparatus (the capsule).
Worm Monster (adattato) _{CR!STALLL (Sketchfab)}	`models/nemici/verme.glb`	CC-BY-4.0	626 kB	ATOMIC PROTOCOL — enemy: the drain worm.
Scary Monster (adattato) _{L. (Sketchfab)}	`models/nemici/curvo.glb`	LicenseRef-Sketchfab-Standard	1145 kB	ATOMIC PROTOCOL — enemy: the lurcher.
SWAMP MONSTER (adattato) _{alexalbinyana (Sketchfab)}	`models/nemici/palude.glb`	CC-BY-4.0	1803 kB	ATOMIC PROTOCOL — enemy: the bog thing.
The Human Deer (adattato) _{ceeleste (Sketchfab)}	`models/nemici/cervo.glb`	CC-BY-4.0	6854 kB	ATOMIC PROTOCOL — enemy: the deer.

Component	Version/File	Licence	Size	Role
6- Stylized- Monster (adattato) _{Lex Wild (Sketchfab)}	`models/nemici/abissale.glb`	CC-BY-4.0	3470 kB	ATOMIC PROTOCOL — enemy: the abyssal.
Monster (adattato) _{itszyx123456789 (Sketchfab)}	`models/nemici/mostro.glb`	CC-BY-4.0	2108 kB	ATOMIC PROTOCOL — enemy: the crystalline.
CREATURE MONSTER 3D BY Oscar Creativo (adattato) _{OSCAR CREATIVO (Sketchfab)}	`models/nemici/bruto.glb`	CC-BY-4.0	5305 kB	ATOMIC PROTOCOL — enemy: the brute.
hands animation (modello 3D, adattato) _{hugo colauto (Sketchfab)}	`models/mani-atomic.glb`	LicenseRef-Sketchfab-Standard	2059 kB	The first-person hands, with the skeleton and the animation the game derives its poses from.
Modern Scientist in a Lab Coat (ritaglio) _{restore50 (Sketchfab)}	`models/manica-atomic.glb`	CC-BY-4.0	275 kB	The first-person lab-coat sleeve, cropped from the scan.
Capsule Gun (modello 3D) _{sc8di (Sketchfab)}	`models/pistola-atomic.glb`	CC-BY-4.0	1366 kB	The capsule gun held in first person.

Digests

File	SHA-256
`RDKit_minimal.js`	`58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9`
`RDKit_minimal.wasm`	`e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8`
`three.min.js`	`9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2`
`three_bloom.js`	`42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090`
`skeleton_utils.js`	`cc2e2e1a3183ab55169d185565f86f43f80c6dfafe73610d991e741ea990ac1a`
`gltf_loader.js`	`5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6`
`3Dmol-min.js`	`06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff`
`smiles-drawer.min.js`	`955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2`
`vendor/pdfjs/pdf.min.mjs`	`bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259`
`vendor/pdfjs/pdf.worker.min.mjs`	`145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddaaec6`
`lib/sql-wasm.js`	`77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f`
`lib/sql-wasm.wasm`	`438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed`
`lib/dimuon.js`	`61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d`
`models/braccia-atomic.glb`	`3e72b0067366ba4906e8cb3d9077e3b800656705bfcdef5b0bb41339c0b765ff`

File	SHA-256
models/mano-robotica-atomic.glb	3cadfde4777f5f609fb9fea7990f35ddc8f82d19fad4c60b8fc18a563ba7e5c6
models/mappe/small.glb	932e21a12c772ef3c36290fac8974a30b37b62e2000f7f1f26cb812a0572a7ab
models/mappe/scifi1.glb	f3e4434820cda4a2ebec5804749cd0a6278aa0cda6a19b74ec14623affb3dc5e
models/mappe/flynn.glb	d9c32659b5a75db09e88e97aeef645f46d30caee24522fc3a18b5048fa56911a
models/mappe/underground.glb	8cae8085cae16d6b496fe93a00eef7d687349492e3d74c7af1aaab407085fa75
models/mappe/overgrown.glb	c521f06e1da35eda66d585a0e6ab5b87a4da504d4b4fb09048518bf11683f403
models/mappe/reattore.glb	58f200ac13b072d40ee45bdd6c725ddd06591502e53825d60699862fc79a3e01
models/mappe/capsula.glb	be1ae03a8aa0c8cd160d517620985f83a515a2f05fc3d062aab4303496befb02
models/nemici/verme.glb	4b35e41fd47e554733f593f671b35b51765367ef2f7522bda4316ddef0ee02d8
models/nemici/curvo.glb	a1fb198ba5806fd97feb45ec4ca5d42a065420ce2bdb0c331699609845218577
models/nemici/palude.glb	83668a9192e57d1183f9086ac974b4e87af8ecdc57b57c13015ec90d2f5da533
models/nemici/cervo.glb	ecece2366754baf17c400ac4265d65ad7daa166a40fc911eece6d3641403291d
models/nemici/abissale.glb	81b701fba1f0a59ce10f57c45128b172aeb1d6f759a51b9b6903f548c14eee2e
models/nemici/mostro.glb	c669349d5562013e251a76fdb48d0e180e3f997e95c07d1064c8697368d0d4e3
models/nemici/bruto.glb	a995a10428f05dd0074570980a72c9a5f04606deb90a58382a8f3e5e3d301ae1
models/mani-atomic.glb	d27b4ea1689e677e9406944abaecd95d524ca7693b2e4b62aec6c906f46ece56
models/manica-atomic.glb	0034162e3fc772d3afb11c2df1a66222ae681ad7f404bcec59977d6557f14b98
models/pistola-atomic.glb	4e109a4ab7e2794b139a5c9414151b376b7a237baab5d9f183f98d8e9ac65f7f

2. Code written for this project

File	Size	Role
index.html	4671 kB	Main application: interface, teaching sections, chemical data.
bsi-ai-hub.js	422 kB	«Spectra» AI agent: agentic loop, tools, provider management.
bsi-spettri.js	35 kB	IR/NMR spectral prediction engine over the molecular graph.
bsi-cheminfo.js	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
bsi-pretsch.js	37 kB	The NMR estimation tables transcribed in full from Pretsch–Bühlmann–Badertscher, 4th ed.: ^{13}C and ^{1}H benzenes, ethylenes, alkanes, alkynes, ^{13}C aliphatics, steric corrections, ^{1}J(C,H). Numbers only, no code.
bsi-nmr.js	57 kB	Per-atom ^{1}H and ^{13}C NMR prediction: applies the bsi-pretsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
bsi-geom3d.js	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.

File	Size	Role
`bsi-nmr2d.js`	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
`bsi-moldraw.js`	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
`bsi-documento.js`	23 kB	Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read.
`bsi-quesito.js`	19 kB	Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js.
`bsi-spettrolettore.js`	104 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
`bsi-mol3d.js`	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.
`bsi-lingue.js`	163 kB	Internationalisation engine: fourteen languages, writing direction included.
`bsi-molingue.js`	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.
`bsi-pannelli-lingua.js`	66 kB	Panels for languages and chemical languages.
`sw.js`	6 kB	Service Worker: offline operation and network strategy.
`rdkit_lab.html`	260 kB	Cheminformatics laboratory.
`astro.html`	2660 kB	Astrochemistry module.
`chimorga.html`	194 kB	Organic chemistry module.
`proxy/spectra-proxy.js`	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
`index.html`	`50c39157bfb20a753ca2de6535259b8e3ca88b7a8beb0d2e21b969aa1b55f6c8`
`bsi-ai-hub.js`	`985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476`
`bsi-spettri.js`	`be757f15a70f884e3374d401596e2dfd18bcc852cafbb4584920e2b2d24eb3c9`
`bsi-cheminfo.js`	`59a3111b3b9436add5376fde1be6f90ded3dc08c095cebdc04da50c7f8268bbc`
`bsi-pretsch.js`	`27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3`
`bsi-nmr.js`	`881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72`
`bsi-geom3d.js`	`a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054`
`bsi-nmr2d.js`	`a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78`
`bsi-moldraw.js`	`45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e`
`bsi-documento.js`	`0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9`

File	SHA-256
`bsi-quesito.js`	`6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf`
`bsi-spettrolettore.js`	`14daf41305b0a7271c6ff5f7c9078a4e5f0e0149f32460a9608ed1996f810256`
`bsi-mol3d.js`	`93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893`
`bsi-lingue.js`	`5a28570f930be99f9e07c46ed354dab3392805820209adb1eb38e28f2d20f410`
`bsi-molingue.js`	`e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb`
`bsi-pannelli-lingua.js`	`4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93`
`sw.js`	`1cc20757acb2497ff8dcbc6d7aee7aec8089a2d48c1fecb98136cb23b6d1a0d4`
`rdkit_lab.html`	`066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad`
`astro.html`	`3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d`
`chimorga.html`	`047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d`
`proxy/spectra-proxy.js`	`808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962`

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
`playwright-core`	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

> **The application's licence is distinct from those of its components.** The code written for BioSpecInfo is **proprietary** — *All rights reserved* (see `LICENSE`): readable and open to assessment, but copying, reuse and commercial use require the Author's written permission. The permissive licences listed above cover **only** the bundled third-party libraries, and do not extend to the application.

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